Maria Helga Grizelda

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Education

University of Queensland — Brisbane, Australia
Bachelor of Science in Computer Science — August 2025 - August 2026

Universitas Indonesia — Depok, Indonesia
Bachelor of Science in Computer Science — August 2022 - August 2026

Experience

Momofin — Jakarta, Indonesia
Software Developer — September 2024 – December 2024

- Built a cryptography-based document verification system for Momofin, enabling institutions and corporations to issue tamper-proof digital documents and instantly verify their authenticity.
- Owned the backend signing and verification module on a Spring Boot and PostgreSQL stack, designing the hashing and digital signature flow that guarantees document integrity end-to-end.
- Shipped the full system from requirements to deployment in a six-person SCRUM team using test-driven development, design patterns, and iterative sprints with regular progress reports.

Leadership & Activities

BEM Fakultas Ilmu Komputer, Universitas Indonesia — Depok, Indonesia
Deputy of Research and Strategic Action (Kastrat) — April 2023 – January 2024

- Led recruitment for 15+ new division members using structured interviews and written assessments.
- Published research briefs and represented student interests in university-level social and political hearings.

Projects

AI Rapid Sorting of Underwater Ecological Imagery

- Built an AI-assisted annotation platform with FFT that accelerates manual labelling of underwater transect imagery for marine biodiversity assessment, with semantic search and human-in-the-loop review.
- Trained an object detection model on a DINOv3 backbone for marine species detection and image embeddings, integrated SAM 3 as a prompt-based assistant tool, and built a retraining loop that learns from reviewed detections and user annotations.

Shopee Sales Performance Dashboard

- Built a Power BI dashboard with custom DAX measures to track sales, cancellations and shipping costs
- Analysed store performance to uncover insights on growth, shipping costs and avoidable cancellations
- Translated operational data into insights on growth, shipping costs and avoidable cancellations

Song Popularity Prediction with Non-Linear Regression

- Performed exploratory data analysis and data cleaning, using multiple imputation to handle missing values
- Built regression models to predict song popularity from audio features, capturing non-linear relationships
- Validated model performance on held-out data and interpreted feature effects on popularity

Food Group and Nutrient Prediction on the Australian Food Composition Database

- Engineered an end-to-end data preparation and modelling pipeline to predict food groups and nutrient values
- Benchmarked multiple supervised learning models to identify the best-performing approach
- Evaluated PCA for dimensionality reduction and its impact on model performance

Skills

Programming Languages: Python, Java, JavaScript, SQL, R, Tableau, Power BI
Frameworks & Tools: FastAPI, Django, Spring Boot, Scikit-Learn, Next.js, PyTorch, Docker, Kubernetes, Git, GCP
Languages: English (Fluent), Indonesian (Native), Japanese (Beginner)